

RAMDEOBABA UNIVERSITY



Department of Computer Science and Engineering

PROJECT SYNOPSIS

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RENTIFY

One-Stop Rental Assistance Platform

Submitted by:

Prisha Shiwankar (Roll No.: 52)

Chetna Deshmukh (Roll No.:08)

Zoya Ghachi (Roll No.: 27)

Naman Kawadkar (Roll No.: 07)

Under the Guidance of:

Dr. Purshottam Assudani, Guide]

Date:

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Formatting Guidelines

Parameter	Specification
Font	Times New Roman
Title	16 pt, Bold, Uppercase
Section headings	14 pt, Bold
Body text	12 pt, Regular
References	10 pt
Line spacing	1.5
Margins	1 inch, all sides
Citation style	IEEE (numbered)
Maximum length	4 pages (this page + 3)

1. Background and Motivation

India's rental housing market has expanded rapidly with rising urban migration, student mobility, and job-driven relocation, yet the process of finding a rental home remains fragmented and largely informal. Students moving to a new city, working professionals relocating for jobs, and migrant families searching for affordable housing typically rely on a patchwork of sources, such as local brokers, word-of-mouth referrals, generic classifieds, and multiple listing apps, none of which offer a single, trustworthy view of the rental market. Listings are frequently outdated, duplicated across platforms, or deliberately misleading, and renters have little way to verify a property, an owner's identity, or the legitimacy of a listing before making a payment or signing an agreement. Beyond the search itself, renters need guidance that current apps rarely provide, such as information on nearby schools, hospitals, transport, and markets, awareness of common rental scams, clarity on legal rights and documentation, and, where applicable, details of government housing assistance schemes. This absence of a unified, trustworthy source of rental information creates anxiety, wasted time, and, in the worst cases, financial loss for renters who fall victim to fraudulent listings. Existing property portals largely optimize for lead generation and advertising revenue rather than renter welfare, leaving a clear gap for a platform designed specifically around trust, transparency, and assistance. This project responds to that gap by proposing a Smart Rental Assistance Platform that verifies listings, centralizes decision-relevant information, and actively helps renters avoid scams, addressing a real and growing need across student, professional, and migrant populations.

2. Problem Statement

Existing rental platforms in India function primarily as listing aggregators: they display large volumes of properties but rarely verify ownership, authenticity, or accuracy of the information provided. Renters must independently cross-check details, negotiate with unverified brokers, and assess safety and legality on their own, often after wasting significant time and, in scam cases, money. Critical decision-relevant information, such as proximity to essential services, area safety, legal rent-agreement norms, and applicable government housing schemes, is scattered across unrelated websites and government portals rather than integrated with the listing itself. Consequently, renters, especially students, migrants, and first-time tenants unfamiliar with a new city, face high search friction and exposure to fraud. No single platform

in the current market combines verified rental listings with contextual, location-aware assistance and structured scam-awareness support. This project proposes Rentify to directly address this integration gap.

3. Objectives

- Provide verified rental listings through a structured owner-verification and document-checking workflow.
- Reduce exposure to rental scams by flagging suspicious listings using rule-based and pattern-based checks.
- Deliver location-aware information on nearby essential amenities, including schools, hospitals, transport, and markets, for each listing.
- Offer accessible rental guidance covering tenant legal rights, standard agreement clauses, and applicable government housing schemes.
- Improve renter decision-making through a transparent, filterable property recommendation module.
- Build an easy-to-use, mobile-responsive platform accessible to Customer, Vendor, Admin, and Delivery/Verification Partner roles.

4. Literature Review and Research Gap

Recent research on digital rental and real-estate platforms has largely progressed along three separate tracks: recommendation systems that rank properties for users, fraud and fake-listing detection models, and blockchain-based systems for verifying ownership and automating lease agreements. Recommender-focused work, such as session-based and context-aware approaches, improves how listings are ranked but assumes the underlying listing data is already accurate. Fraud-detection studies apply clustering or classification techniques to flag suspicious listings but are typically evaluated on offline datasets rather than deployed as part of an end-to-end rental workflow. Blockchain-based systems strengthen trust in ownership and lease execution but add infrastructure complexity and rarely address the renter's need for contextual information such as nearby amenities, safety, or legal guidance.

Sr. No.	Author(s) & Year	Title / Focus	Methodology / Key Contribution	Limitation / Gap
1	Polohakul et al., 2021	Real Estate Recommendation Approach for Solving the Item Cold-Start Problem	Hybrid session-based and content-based recommender for new listings and new users	Focuses only on recommendation ranking; no verification or scam-detection layer
2	Mohd Amin et al., 2024	Clustering Analysis for Classifying Fake Real Estate Listings	K-means clustering with distance metrics to separate fake from genuine listings	Applied to a static sale dataset; not integrated into a live rental verification workflow
3	Xue et al., 2021	Housing Rental System Based on Blockchain Technology	Alliance blockchain for identity/property authentication and automated lease smart contracts	High infrastructure complexity; limited amenity or legal-guidance support for tenants
4	Nguyen et al., 2023	Design a Recommendation System in Real Estate Investment Based on Context Approach	Content- and session-based filtering combining user profile and browsing context	Targeted at investment property search rather than tenant-focused rental assistance
5	Bhanushali et al., 2020	Blockchain to Prevent Fraudulent Activities: Buying and Selling Property Using Blockchain	Smart-contract based ownership verification to reduce transaction fraud	Designed for sale transactions; does not address rental-specific scam patterns
6	Levandowski et al., 2012	LARS: A Location-Aware Recommender System	Spatial indexing that combines user location with item and user ratings	General-purpose location recommender, not adapted to rental amenity-based search

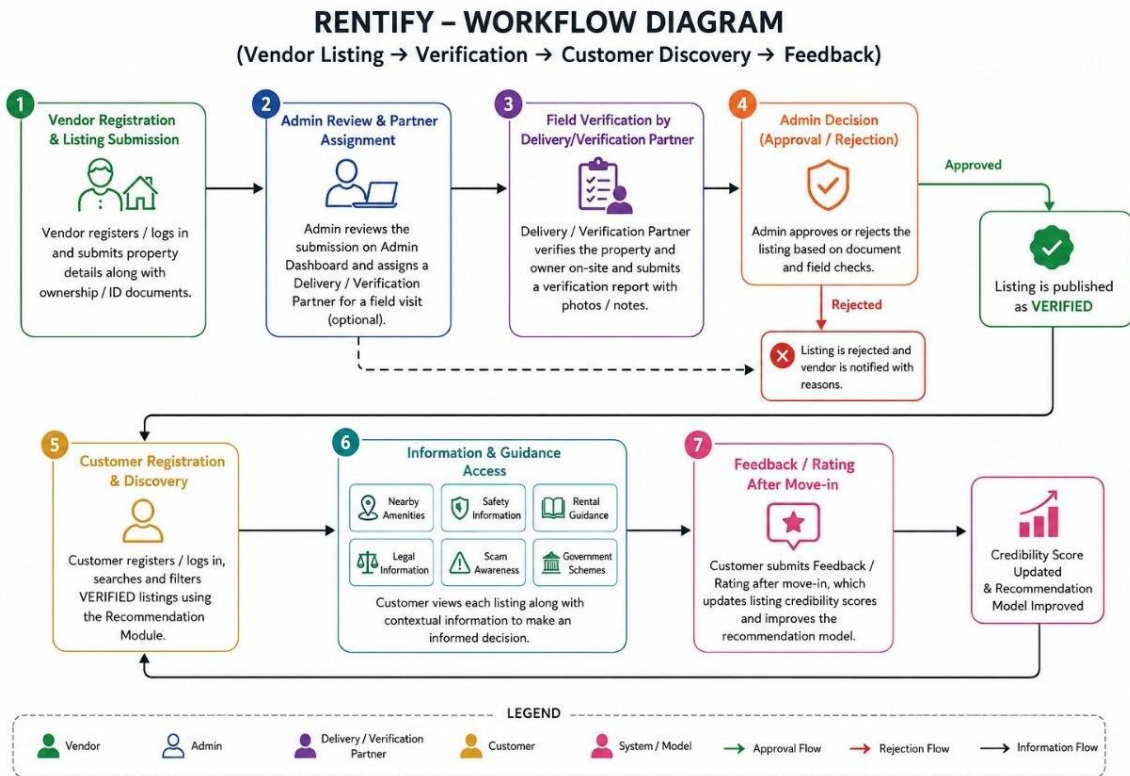
Research Gap:

Existing research treats verification, recommendation, and rental guidance as isolated problems, and no reviewed system combines all three within a single renter-facing platform. None of the surveyed works integrate owner/listing verification with location-aware amenity information, scam-awareness education, and government housing-scheme guidance in one interface. Rentify is differentiated by unifying a lightweight verification workflow, a context-aware recommendation module, and structured rental-assistance content (safety, legal, and scam-awareness information) into a single platform purpose-built for the Indian rental market, rather than adapting a general real-estate or blockchain system to the renter's actual decision-making needs.

5. Proposed Methodology / Plan of Work

The platform follows a modular architecture built around four user roles: Customer, Vendor, Admin, and Delivery/Verification Partner. User Registration and Authentication use OTP/email-based sign-up with role-based access control, ensuring each role sees only the workflows relevant to it. In the Rental Listing Module, Vendors submit property details along with ownership proof and identity documents, which are queued for review rather than published immediately. The Property Verification Module routes each submission to an Admin, who performs a document check and can assign a Delivery/Verification Partner for an optional physical verification visit, after which the listing is marked Verified or flagged. The Recommendation Module ranks verified listings for each Customer using a content-based filter (budget, location, property type) combined with collaborative signals from similar users' interactions, addressing the item cold-start problem highlighted in the reviewed literature. The Rental Assistance Module aggregates, for each listing, Nearby Amenities fetched through a maps API, Safety Information drawn from publicly available civic data, Scam Awareness content highlighting common red flags in rental fraud, and Legal Guidance summarizing standard rent-agreement clauses and tenant rights, with links to applicable Government Housing Schemes where relevant. A Feedback System allows Customers to rate listings and Vendors after move-in, feeding back into listing credibility scores and the recommendation model. The system is built iteratively: Vendor and Admin workflows first, to establish a base of verified listings, followed by Customer-facing search, recommendation and assistance features, and finally the feedback loop that closes the trust cycle. The end-to-end flow is summarized below.

Workflow Diagram (Vendor Listing → Verification → Customer Discovery → Feedback):



6. Technology, Tools and Platforms

Programming Languages: JavaScript, Python. Frontend: React.js with Tailwind CSS. Backend: Node.js with Express.js. Database: MongoDB (listings, users) with PostgreSQL for structured verification records. Frameworks: REST APIs. Maps: Google Maps/OpenStreetMap API for amenity and geolocation data. Authentication: Firebase Auth with OTP verification. Cloud Platform: AWS (EC2, S3) or equivalent free-tier cloud service. Version Control: Git and GitHub. Development Tools: VS Code, Postman. Testing Tools: Jest, Postman for API testing. Deployment Platform: Vercel/Render for frontend and backend hosting.

7. Expected Outcomes, Deliverables and Functional Specifications

The project will deliver a working web-based platform, Rentify, comprising Customer, Vendor, Admin, and Delivery/Verification Partner modules, a verified-listing database, a location-aware recommendation engine, and an integrated rental-assistance section covering amenities, safety, legal guidance, scam awareness, and government scheme information. Deliverables include the deployed application, source code repository, verification workflow

documentation, and a test report. The expected impact is a measurable reduction in time-to-find-verified-housing and reduced exposure to fraudulent listings for students, professionals, and migrant renters. Future enhancements include a mobile application, AI-based document forgery detection, chatbot-based rental guidance, and multilingual support for wider accessibility across Indian states.

8. Project Scope

The current scope covers verified rental listings, owner and document verification, location-based amenity and safety information, rental guidance content, and a feedback-driven recommendation system for residential rentals within a single city or region as a pilot. Target users are students, working professionals, migrants, and families seeking rented accommodation, along with property owners and brokers seeking verified visibility. Current limitations include manual elements in document verification and a text-based (non-multilingual) interface at the pilot stage. The system is designed to scale to multiple cities, integrate AI-based automated document and image verification, add chatbot-driven legal and scam-awareness assistance, and extend to a native mobile application as future scalability directions.

9. Project Timeline

The project is planned across a twelve-week schedule beginning with literature review and requirement analysis, followed by system design, iterative module development, testing and evaluation, and concluding with documentation and final submission, as summarized below.

Phase	Duration	Milestone / Deliverable
Literature review & requirement analysis	Week 1 - 2	Finalized problem statement and objectives
Design & methodology finalisation	Week 3 - 4	System architecture and database schema
Implementation/development	Week 5 - 9	Functional modules: listing, verification, recommendation, assistance
Testing & evaluation	Week 10 - 11	Test report and bug-fixed build
Documentation & final submission	Week 12	Final report, demo, and project submission

10. References

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- [6] D. Bhanushali, A. Koul, S. Sharma, and B. Shaikh, "Blockchain to Prevent Fraudulent Activities: Buying and Selling Property Using Blockchain," in Proc. 2020 Int. Conf. Inventive Computation Technologies (ICICT), IEEE, 2020, pp. 705-709.
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- [8] J. J. Levandoski, M. Sarwat, A. Eldawy, and M. F. Mokbel, "LARS: A Location-Aware Recommender System," in Proc. 2012 IEEE 28th Int. Conf. Data Engineering (ICDE), 2012, pp. 450-461.

Group Member Details

If a student belongs to another branch, specify the branch name beside the student's name.

Roll No.	Name and Signature of Student
52	Prisha Shiwankar
08	Chetna Deshmukh
27	Zoya Ghachi
07	Naman Kawadkar

Approved by:

Dr. Purshottam Assudani

Project Co-Coordinator: Prof. Tintu Philip

Project Coordinator: Dr. Sharvari Deshmukh

Associate Head of Department: Dr. Preeti Veditel

Dean CSE: Dr. Padma Adane